

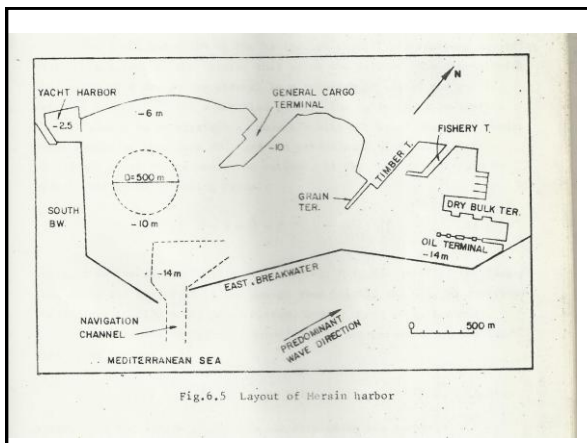
Harbor (Port)

- Coastal installation for sheltering and providing maintenance for ships and for loading and unloading cargo and passengers.



Harbor or Port?

- Harbor is an area of water affording a natural or artificial shelter for ships.
- Harbor facilities: channels, anchorage and mooring basins, dry docks, ship repair plants
- Port is a harbor plus terminal and transfer facilities.
- Port facilities: piers, warehouses, cargo handling equipments, railroads etc.



Harbor or Port?



ÇANDARLI PORT



ÇANDARLI PORT

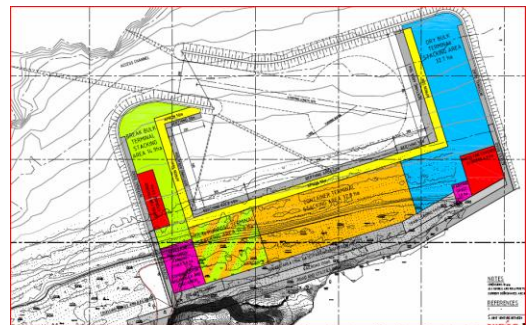


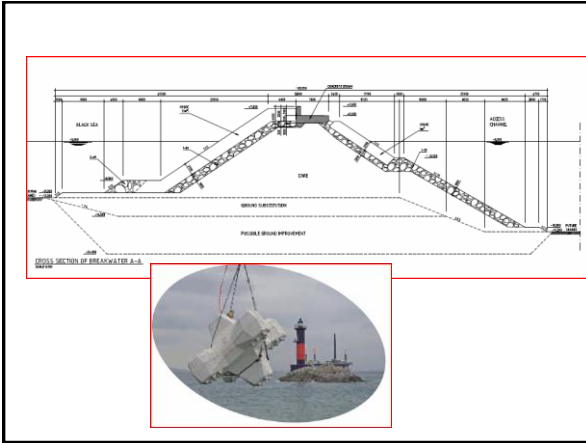
- 500m rubble mound BW
- 1000m pile (L=90m, D=1.78m)

Filyos Port

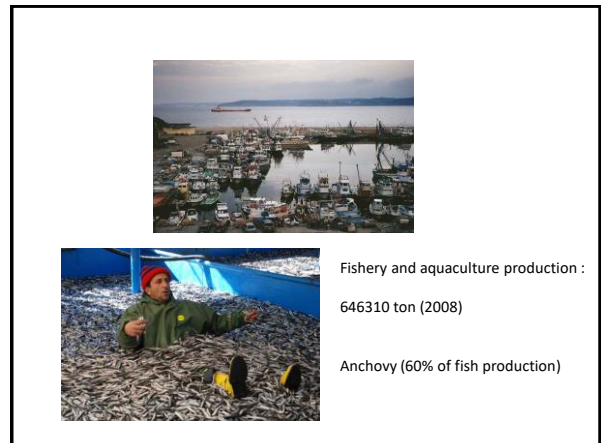
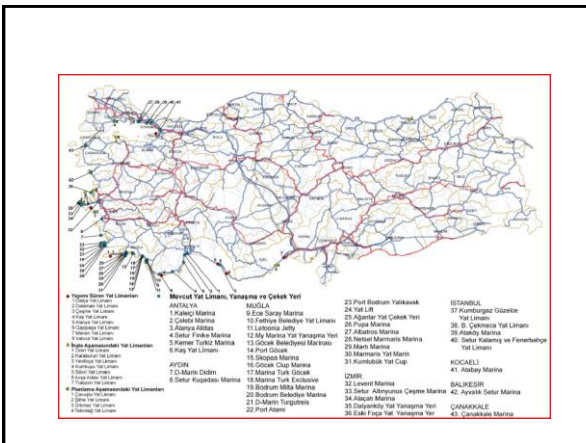


Filyos Port





Marinas



Fishery and aquaculture production :

646310 ton (2008)

Anchovy (60% of fish production)

- Ulaştırma Kıyı Yapıları Master Planı

TÜRKİYE TOPLAMI	2030 Tahmini (Ort)	Mevcut Kapasite	Kapasite Farkı
Konteyner (TEU)	30,129,234	9,185,300	20,943,934
Genel ve Dökme Yük (Ton)	484,185,854	253,765,810	230,420,044
Sıvı Kimyasal Yük (Ton)	179,442,035	144,043,000	35,399,035
Ro-Ro Diş. Ticaret (Adet)	1,082,031	1,679,800	-597,769
Ro-Ro Kabotaj (Adet)	1,714,316	250,000	1,464,316
Otomobil (Adet)	5,938,210	1,989,230	3,948,980

- Önümüzdeki 15 yıl içerisinde ülkemizdeki ihtiyaç:
- 10 adet konteyner limanı (her biri en az iki milyon TEU)
- 23 adet genel kargo limanı (her biri 10 milyon ton elleçleme kapasitesinde)

Türkiye Geneli Yat Talep Tahmini

(Turizm Kıyı Yapıları Master Planı,
Ulaştırma Denizcilik ve Haberleşme Bakanlığı, 2010)

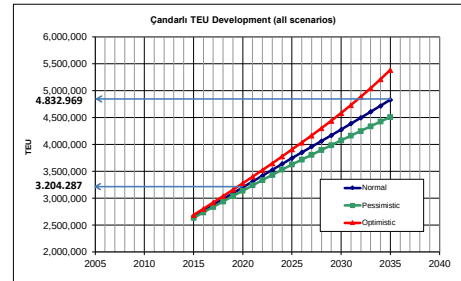
	2015	2020	2025	2030
	17,648	20,791	23,880	26,811
	23,281	34,114	48,144	65,768

- Türkiye'de 20'nin üzerinde yat limanı projesi

Harbor Planning and Layout

- Site selection
- Estimating the harbor size and facilities
 - Projection of port traffic
 - Type and size of design vessels
 - Berth requirement
- General harbor layout

Traffic Forecast Results



TEU=Twenty-Foot Equivalent Unit
=34 m³

Demand
Forecast

Site Selection

- Selection of the region
- Selection of the local site

Site Selection

- Selection of the region
 - Foreland studies:
 - Near an important seaway
 - Intersection of seaway and highway



Transshipment

Origin-Destination Potential for Çandarlı:

•Syria	➡	Italy
•Italy	➡	Israel
•Greece	➡	Spain
•Bulgaria	➡	Egypt
•Romenia	➡	Israel

Black Sea countries with developing trade potentials will specifically involve in transshipment loop for the new Çandarlı port.

Source:
UMA

Site Selection

- Selection of the region
 - Hinterland studies
 - Hinterland: inland areas using the port
 - Developing cities with their industrial or agricultural products
 - Good railway and highway
 - Not very high and hilly



Site Selection

- Selection of the local site
 - Requires site investigations to determine the geotechnical and the coastal hydraulic conditions
 - Geotechnical studies
 - Water depth
 - Waves and water levels
 - Currents and littoral drift
 - Pollution
 - Availability of building material

Site Selection

Soil Condition:

- Ask to local people
- Check previously constructed structures near the site
- Soil investigations (driving pile, make borings, lab tests, SPT tests)
- Favorable soil condition for the construction of structures and terminal facilities
- if the soil is soft clay, soil improvement or use pile until you reach to rock so expensive
- If it is rock, underwater excavation of rock is very expensive, avoid if possible

Site Selection

Water Depth:

- Bathymetric map should be drawn
- Selected coastal location should involve the least amount of dredging.
- If the water depth is great, then the construction of breakwaters and piers is expensive.
- If the water is shallow then dredging becomes expensive.

Site Selection

Waves:

- A naturally protected location is ideal
- An open sea harbor will need longer breakwaters.
- Wave measurements or long term hindcasting studies



Site Selection

Currents and littoral drift:

- Heavy littoral drift along the shore is undesired as it may fill the harbor entrance (silting)
- A Harbour should not be located at the mouth of a river, if the river is not navigable

Site Selection

Environmental impact:

- Environmental pollution (waste disposal of industries around the port and disposal from ships)
- Esthetical aspects

Site Selection

Availability of building materials:

- Rock, timber, steel in the vicinity of the site should be investigated.

Site Selection

- For the selection of the site, it may be impossible to fulfill all of the conditions.

Non-technical factors:

- National policy
- Political reasons (view of existing government)
- Military reasons

HW1

- Make a group (2 people)
- The region will be given to you
- Investigate the harbor type necessary in that region
- Select a good location